Name :- Sana Munir Alam
Roll # :- 24K-0573
Class :- BCS-4E

AI - Assignment 1

Date: 16-02-2026

NASA's CASPER

Q1a) Core Objective:

Continious Activity Scheduling, Planning, Execution and Replanning (CASPER) was designed to enable continious real-time autonomous planning and execution for spacecraft operations. Unlike traditional batch planners that generate complete plans before execution. CASPER maintains and modifies plans dynamically during execution, allowing spacecraft to respond to unexpected event anomalies, and new science oppurtunities without waiting for ground control intervention.

Theoretical Foundation:

1- Iterative Repair Planning :- Instead of planning from scratch, CASPER modifies existing plans by identifying conflicts and applying local repairs.

2- Constraint Based Reasoning :- Uses temporal, resource and state constraints to ensure plan vadility.

3- Model-Based Representation: Encodes spacecraft capabilities, limitation and operational rules as formal models

4- Continious Planning Paradigm: Maintains a "current plan" that is constantly updated rather than regenerated.

AI Goals:

CASPER embodies AI's goal of autonomous agency in dynamic environments.

- Reactivity: Responds to changes in seconds rather than hours.
- Robustness: Handles uncertainity and unexpected events.

Date:

- Goal-Directed Behavior: Maintains high-level science and engineering objectives while adapting low level execution.
- Limited Rationality: Produces "satisficing" (good enough) plans under time constraints rather than optimal solutions.

b)

Goals & Constraint INPUT

- Science Goals (targets)
- Resource Constraint
- Engineering Goals

↓

Synchronization Modules

- Queues and serializes update requests

↓

Planner Process (CASPER Core)

1- Plan Database (Current Plan)

1. Activities (task)
2. Temporal Constraint
3. Resource Projection

2- Iterative Repair Engine

1. Conflict Detection
2. Repair Selection
3. Heuristic Search

↓

Execution Module

1. Commit Activities
2. Command Execution
3. Monitor Status

↓

State Estimation Module

- Sensor Data
- Resource Update
- Anamoly Detection

Date:

↓

(feedback loop to Planner)

Role of Each Layer

Layer	Function	Decision-Making Role
Input Layer	Recieves science targets, engineering requirements, resource limits	Defines what needs to be achieved.
Synchronization	Serializes concurrent updates (goals, state changes)	Prevents race conditions in plan updates
Plan Database	Maintains current plan with temporal and resource projection	Stores the "world model" and expected future state
Iterative Repair	Detects conflits (constraint violations) and applies repails	Core Intelligence: Transform invalid plans int valid ones
Execution	Commits activities to execution issues commands, monitors status	Bridges planning abstraction with physical actions.
State Estimation	Processes sensor data, updates resource levels, detect anomalies	Grounds the plan in physical reality.

Date:

c) • Constraint Management System:
It maintains temporal, resource, and state constraints to ensure spacecrapt remain valid. The constraint system forms the logical backbone of CASPER. It enables systematic detection of inconsistencie before execution. As all activities are encoded declaratively, conflicts are automatically detectable.

Strength	Limitation
• High internal consistency and safety assurance	• Highly dependent on model accuracy
• Prevents resource violations	• If space craft models are incomplete intelligent behavior degrades

• Iterative Repair Algorithm
It detects conflicts and applies local modifications to restore plan validity. It interacts with constraint by using constraint violation as triggers. Selects repair operators (move, insert, delete). Uses heuristics to guide search.
It enables fast response compared to planning from scratch, support real time autonomy. maintains plan continuity. Computationally efficient for onboard systems. Produces satisficing rather than optimal plans.
The repair engine creates adaptive behavior, but its intelligence is procedural rather than strategic.

• Temporal Reasoning System
Maintains time windows, durations, and active dependencies. It interacts with repair engine, enables look ahead projection, detects future conflicts before execution. Prevents cascading failures, and provides predictive intelligence rather than reactive only

Date:

behavior. However complexity increases with mission duration, computational overhead grows with plan size.

• Model Based Domain Representation
It encodes space craft subsystems, resources (power, data storage, fuel), and operational modes. All reasoning depends on this model. It enables general rule-based planning instead of fixed scripts. Allows flexibility across different mission (eg: EO-1, IPEX). requires extensive manual modeling. Poor transferability across radically different domains.

• Execution and State estimation Modules.
It executes commands, monitor spacecraft status and feed updates back to planner. Enables continious planning paradign as it creates a closed feedback loop. It reduces dependence on ground control and real-time feedback allows dynamic adaption.

Overall the intelligence remains algorithmic and domain-bound as the system lacks learning capability, abstraction beyond modeled knowledge, and self-generated objectives.

d) Research Impact
NASA's CASPER pioneered the continious planning paradigm, marking a shift from traditional batch planning to anytime dynamic planning under uncertainity. This work significantly influenced later autonomous planning systems such as Autonomous Sciencecraft Experiment (ASE), CLEaR and multi-agent rover coordination framework. It demonstrated that onboard delibrative autonomy is feasible and

Date:

reliable for missions where communication delays make ground-based control impractical, particularly in deep-space exploration.

~~Mission~~ Industry Impact:
CASPER was successfully deployed on Earth Observing-1, where it enabled autonomous detection and prioritization of scientifically valuable events such as volcanic activity (Mt. Erebus, 2004), floods and cryospheric changes. CASPER also contributed to automation within the Deep Space Network by improving antenna tracking and scheduling efficiency. Collectively, these deployment reduced operational costs by minimizing continious human supervision.

Societal Impact.
Enabling real-time autonomous decision making, CASPER increases scientific return through the capture of transient phenomena that would otherwise be missed due to communication latency. It also proved that autonomous AI could safely control expensive national assets in real-time.

Intelligence Classification :- NARROW INTELLIGENCE
Reason :- CASPER represents sophisticated narrow intelligence, it While it exhibits intelligent behavior such as planning, adaption and conflict resolution, it's operation is restricted to well defined space-craft control domains. It lacks cross-domain generalization, transfer learning capability, semantic understanding self-directed goal formation and common-sense reasoning CASPER doesn't understand space science, instead it manipulates symbolic constraint representation using pre-defined models. Thus.

Date:

its intelligence arises from algorithmic sophistication rather than general cognitive ability.

e) Technical limitiation.

Limitation	Impact On Reliability
• Computational Constraints	Onboard processor limit plan complexity and repair speed.
• Model Dependency	Requires accurate spacecraft models as model erras cause plan failure
• Local Optimal	Iterative repair may find ~~sta~~ satisficing but not optimal solutions.
• Verification Difficulty	Continious modification makes exhaustive testing impossible.

Ethical Limitation

- Opacity :- While CASPER's decision are based on explicit constraints, the interaction of repairs can be unpredictable behavior in complex plan.
- Accountability :- If autonomous spacecraft damages itself or misses critical science, responsibility is distributed between engineers, scientist and the algorithms.
- Risk Acceptance :- Autonomy reduces human oversight which is acceptable for robots, but precedent for higher stake applications.

Date:

Practical Limitation

- Engineering Cost:- Extensive modelling required for each new spacecraft (slow build releases due to very consuming testing times).
- Reduced Flexibility:- Trade-off between response time and observatis flexibility, more autonomy means less ability to accomodate unusual science requests.

These limitation restrict CASPER to missions where communication delays make ground control impractical. cost of autonomy development is offset by operations savings. Risk tolerance allows for occassional suboptimal decisions.

f) The improvements can include

→ Adaptive Intelligence

- Learning-Based Repair: Replace hand coded heuristics with learned repair strategies from historical mission data.
- Meta-Learning: Enable CASPER to adapt it's planning strategy based on mission phase and success/failure pattern.

→ Techinical Improvement

- Hybrid Architecture: Combine CASPER's delibrative planning with deep learning for pattern recognition in Science data.
- Edge Computing: Leverage modern space-qualified processors for more sophisticated reasoning.

→ Generalization Extension:

- Domain Independent Core: Seperate Generic continious planning

Date:

engine from space-craft specific model
- Cross-Mission Learning :- Transfers knowledge from EO-1 to future missions without complete remaking.

Q2) Field :- Health Care

a) One of the most impactful emerging applications of AI in health care is deep learning based diagnostic imaging. This technology uses advanced neural network architectures, particularly "Convolutional Neural Networks (CNNs), to analyze medical images such as X-rays, CT scans, MRI scans, and retinal (fundus) photographs. These systems assist in detecting abnormalities, classifying diseases, segmenting tumors, and in some cases generating automated diagnostic reports.

A notable real world example is "IDx-DR", the first fully autonomous AI system cleared by the US. Food and Drug Administration (FDA) for diagnosting diabetic retinopathy directly from retinal images without requiring a specialist to interpret the results. This milestone demonstrated the trust in AI systems for independent clinical decision report.

In 2025, regulatory data indicate that a significant majority of AI/ML enabled medical devices approved by Food and Drug Administration (FDA) in US are designed for radiology, with roughly 77% of such devices dedicated to imaging tasks.

Recent research and development also emphasize multimodal vision-language AI systems that combine visual and textual reasoning for enhanced diagnostic and report generation.

Date:

b) Working Mechanism

AI powered diagnostic imaging systems typically follow a structured computational pipeline.

- First, medical images are acquired in standarized formats such as DICOM. These images may include X-rays CT scans, MRI images or retinal photograph.
- Second, preprocessing technique are applied. These include image normalization, noise reduction, resizing and augmentation (during training). Preprocessing ensures that the model recieves standarized and high quality input data.
- Third, the processed image is passed through a deep neural network, typically a CNN architecture such as ResNet, U-Net, or Vision Transformers. The network extracts hierarchical features from the image, progressing from simple patterns (edges and textures) to complex anatomical structures.
  The system then performs classification (disease present/absent), localization (identifying abnormal regions), or segmentation (pixel-level tumor boundaries).
- Finally, the system produces outputs such as diagnostic labels, probability/confidence scores, bounding boxes around lesions, or segmentation masks.
- In advanced systems, multimodal models combine image features with clinical data to generate structured radiology report.

Diagnostic Performance

Clinical trials and meta analyses demonstrate strong performance across multiple applications. For eg IDx-DR achieved sensitivity of approximately 87% and specificity of approximately 90% for

Date:

detecting referable diabetic retinopathy. Broader meta-analyses of AI based screening systems report pooled sensitivities near 95% and specificities above 90% in controlled settings. Similar performance levels have been reported in mammography and chest X-ray analysis.

Applications

AI powered imagings systems are applied in several clinical domains

- Diabetic Retinopathy Screening: Autonomous detection from retinal photographs in primary care settings
- Lung Nodule Detection: Early identification of suspicious nodules in CT scans.
- Mammography Analysis: Breast cancer screening and reduction of false positives.
- Emergency Imaging Triage:- Rapid prioritization of stroke or trauma cases.

These application improve early detection, accelerate clinical work flows, and support decision-making in high-volume health care enviroments.

c) Advantages

It offers several important benefits

- First, it significantly increases diagnostic speed, often processing images within seconds compared to minutes or hours required for manual interpretation.
- Second, it achieves high accuracy in many screening tasks, with sensitivity and specificity often exceeding 90%.
- Third, AI systems provide scalability and consistency. Unlike

Date:

human clinicians, they don't experiance fatigue and can process large volumes of images continiously.

Challanges

- One major is "generalization across populations." Models trained on data from one hospital or demographic group may perform poorly when deployed in different settings due to domain shifts.
- Another challenge is "data heterogeneity", as variations in imaging devices, protocols, and image quality can reduce model robustness.
- Lastly, poor quality or ungradable images can significantly impact diagnostic accuracy. ~~These~~ Plus insufficient training data for uncommon conditions.

These challanges highlight the importance of robust validation before deployment.

Ethical Issues

- Bias and fairness are critical issues. If minority population are underrepresented in training data. AI systems may demonstrate unequal diagnostic performance across demographic grp, potentially worsening health care disparities.
- Accountability remains unclear when AI systems make incorrect diagnoses. Determining whether responsibility lies within the developer, healthcare provider, or institution is legally complex.
- Issues of informed consent arise when patients are unaware that AI systems contributed to their diagnoses

Date:

d) Technical Approvement

- Explainable AI: use of attention maps, concept activation vector (CAV), counterfactual explanations. This way clinician can se "where AI is looking" and why.
- Federated Learning: Train on distributed hospital data without centralizing. This preserve privace while using diverse datasets.
- Uncertainity Quantification:- Includes Bayesian neural networks and ensemble model, can provide confidence interval alongside predictions, enabling safer clinical decision making.

Clinical Integration Development

- Future systems may adopt multimodal integration, combining imaging data with genetic information, electronic health records, and clinical notes to provide more comprehensive diagnostic support.
- Continious learning frameworks, where AI system update safely with new data under controlled supervision, avoiding performance degradation while maintaining regulatory compliance.

Q3a) AI Based Resume Screening System (PEAS)

Component	Description
• Performance Measure	• Accuracy of candidate job matching (precision, recall) • Reduction in time-to-hire • Diversity and fairness metrics. • Candidate staisfaction scores • Cost per hire reduction • False positive/negative rates

Date:

Environment	• Job descriptions and requirements. • Candidate resumes / CV's • Application Tracking Systems • Company hiring policies and constraints • Labor market condition • Recruiter and hiring manager preferences • Historical Hiring data.
Actuator	• Resume ranking / scoring outputs • Automated rejection / acceptance notifications. • Candidate shortlisting for interviews • Feedback generation for candidates • Reports for HR analytics. • Integration with scheduling systems for intervias
Sensors	• Resume parsers (extract text) • Natural language processing engines • Job decription analyzers • Historical performance databases • Assesment test results • Video interview analysis • Background Check data.

Date:

ii Flood Detection AI (PEAS)

Component	Description
Performance Measure	• Detection accuracy (true positive rate, false alarm rate) • Warning time provided before flood event • Precision of flood depth. • Lives saved and property damage prevented.
Environment	• River basins, watersheds, and drainage systems • weather system and precipitation patterns • Topography and land use data • Soil saturation levels • Urban infrastructure • Historical flood record • Tidal patterns and storm surges.
Actuator	• Early warning alerts (SMS, sirens, mobile app) • Emergency Notification Systems • Flood barrier/gate controls automated • Traffic management system • Evacuation recommendation system
Sensors	• Rain gauges, and river level sensors • Weather satellites and radar. • Soil moisture sensor • Seismic sensors.

Date:

3b) Resume AI (Environment Classification Table)

Classification	Reasoning
Partially Observable	The system can't fully observe candidate's true capabilities, cultural fit, or future performance.
Single Agent	Primarily operates as a single decision-maker (the AI system). While human (recruiters) are involved they act as supervisors rather than competing agents in the environment.
Stochastic	Outcome of hiring decision is uncertain. As same resume quality does not gurantee same job performance, Candidate success depends on many unpredictable factors.
Episodic	Each resume screening is an independent episode with no direct dependency on previous decisions.
Static	Job market evolves slowly. and resume don't change during processing.
Discreete	States are discreete (accept / reject / short list), inputs are discreete documents, and time steps are discreete per application.

Date:

Flood Detection AI

Partial Observability	Can't fully observe all watershed condition, underground water flows or future weather.
Multi-agent	Multiple sensors, weather model and govt must co-ordinate actions.
Stochastic	Flood occurence depends on chaotic weather systems, unpredictable human action and random variable in natural processes.
Sequential	Current flood risk depends on previous states (cumulative rainfall, soil saturation, river level). Decisions have temporal dependencies.
Dynamic	Environment change continiously, like rainfall, river levels and terrain conditions evolve in real time while agent is delibrating
Continious	Water levels, rainfall rates, and time are continious variables. Flood propagation is a continious physical process.

Date:

Q4) Implicit Agent: The implicit agent is an Automated Book Retrieval system (ABRS) or a library retrieval robot. This agent is responsiple for navigating a grid based library to locate and fetch a book.

Problem formulation :-

- State Space: A set of tuple respresenting (x, y, h) where (x, y) are the grid co-ordinates of the robot and (h) is a boolean value indicating whether the robot is currently holding a book.
- Initial State: The starting fixed location of the system (x start, y start) with no book held ($h = false$)
- Action :- Possible mores include, move south / north / east / west and Pick up Book. The movement is only possible if the target aisle is not blocked.
- Transition Model:
  - If the action is a movement to an adjacent unblocked cell (x', y'), the state changes from (x, y, h) to (x', y', h) aka new location.
  - If the action is 'Pick up Book' at the book's location eg (x = book_x, y = book_y) the state change from $(x, y, False)$ to $(x, y, True)$.
- Goal Test :- A condition that returns true if the agen is at the specific book's shelf location (x = book_x, y = book_y) and state is $h = True$.
- Path ~~Test~~ cost: Each movement action $(x, y) \rightarrow (x', y')$ has a cost "c" (typically $c = 1$ per grid cell) to represent operational time and energy. The "Pick up Book" action may have

Date:

seperate cost.

Solution :- The solution is an optimal sequence of actions that transforms the initial state into the goal state with the minimum total path cost. Reason being that the system knows the full layout and location, it can use an informed search algorithm like A^* to find the most efficient route around maintainance blockages.

Q5a) Search Strategy : Iterative Deepening Depth First Search.
The drone is using this strategy and it combines the low memory needs of DFS with the competeness of BFS.

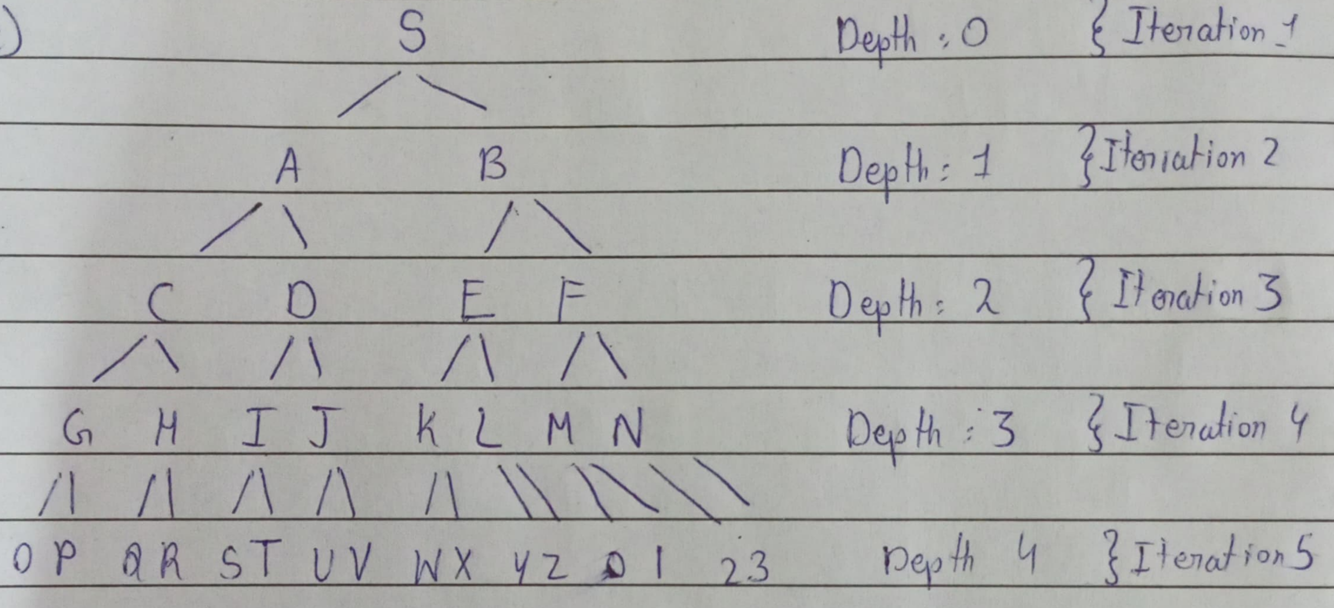
- The drone conducts 2 series of depth limited searches, starting with a limit of 0
- It explores all nodes at the current depth /level using a depth first path.
- If the exist is not found, it returns to the start, increases the depth limit by one, and repeats the entire process again until goal found.

b) Appropriateness of the Strategy

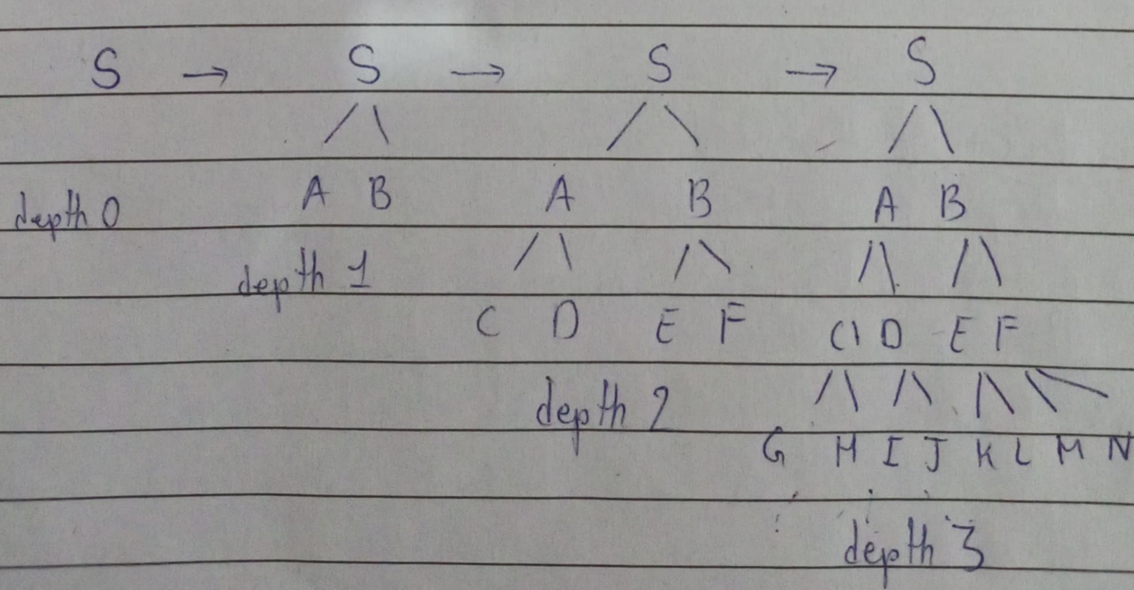
- Limited Memory : IDDFS only needs to store the current path it's exploring. It's space complexity is O(bd) which is far more efficient then storing an entrire frontier like BFS.
- Unknown Depth :- As IDDFS increases the limit incrementally, it is complete. It will eventually find the exist regardless of how deep it is without having to know the complete depth in advance.

- Deep dead ends: Unlike a standard DFS, which could follow a single deep tunnel forever, IDDFS uses a depth cutoff which forces the drone to stop and back track ensuring it doesn't get lost in an infinite or very deep dead end.

c)



at each iteration the depth increases one by one. By the time the drone reaches depth 4 on 5th iteration. This could be the cave system ~~be~~ it explored.



Date:

Q6) DFS may fail if the limit is too small or waste time if too large, and it doesn't gurantee the shortest path.
IDDFS search systemetically increase depth, find the goal without knowing it's depth, gurantee the shortest pain and uses low memory. Therefore IDDFS is more suitable for this maze